

EXERCISE - 8

OS NMAP LIVE HOST DISCOVERY

Aim:

Discover live active host in a network using scans such as ARP, ICMP, TCP and UDP.

Objective:

To understand how Nmap identifies active systems before performing further network scanning.

Software:

Nmap

Try hack Nmap from attackbook computer with network access.

Theory:

Nmap is a free and open-source network scanning tool used to discover live hosts and identify network services. It was created.

* Nmap uses different live hosts using different protocols.

1) ARP Scan - uses ARP scan requests to find host on same subnet.

2) ICMP Scan - uses ICMP request and replies to identify.

3) TCP Scan - uses TCP request and packets to specified ports and check the response.

4) UDP Scan - send UDP packets and can identify an active host through an ICMP port unreachable.

Observation:

* ARP request can discover hosts within the same subnet.

* ARP cannot cross routers because it operates at the link layer.

* An ACK ping generally requires a predefined account.

* UDP ping can identify an active host when an ICMP port unreachable is received.

* The -sn option allows nmap to perform host discovery without conducting a port scan.

Result:

Thus, the live host discovery using Nmap was successfully studied and performed. The experiment helped in understanding how active hosts can be identified before performing further network scans.
